SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 20**
Question Count: 4

Code of Conduct

I J.Rupash [192511170] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Rupash J.

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing

Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Concept Map : Cloud Computing Fundamentals :

Cloud computing

A model of computing that delivers compute, storage, and application via the Internet instead of local machines

- has characteristics
- Provides benefits
- faces limitations
- is based on key concepts

Characteristics

* on demand access
* Resource pooling
* Elasticity
* scalability
* Measured service

Benefits

* Cost efficiency
* Flexibility
* High availability
* Global access
* Faster deployment

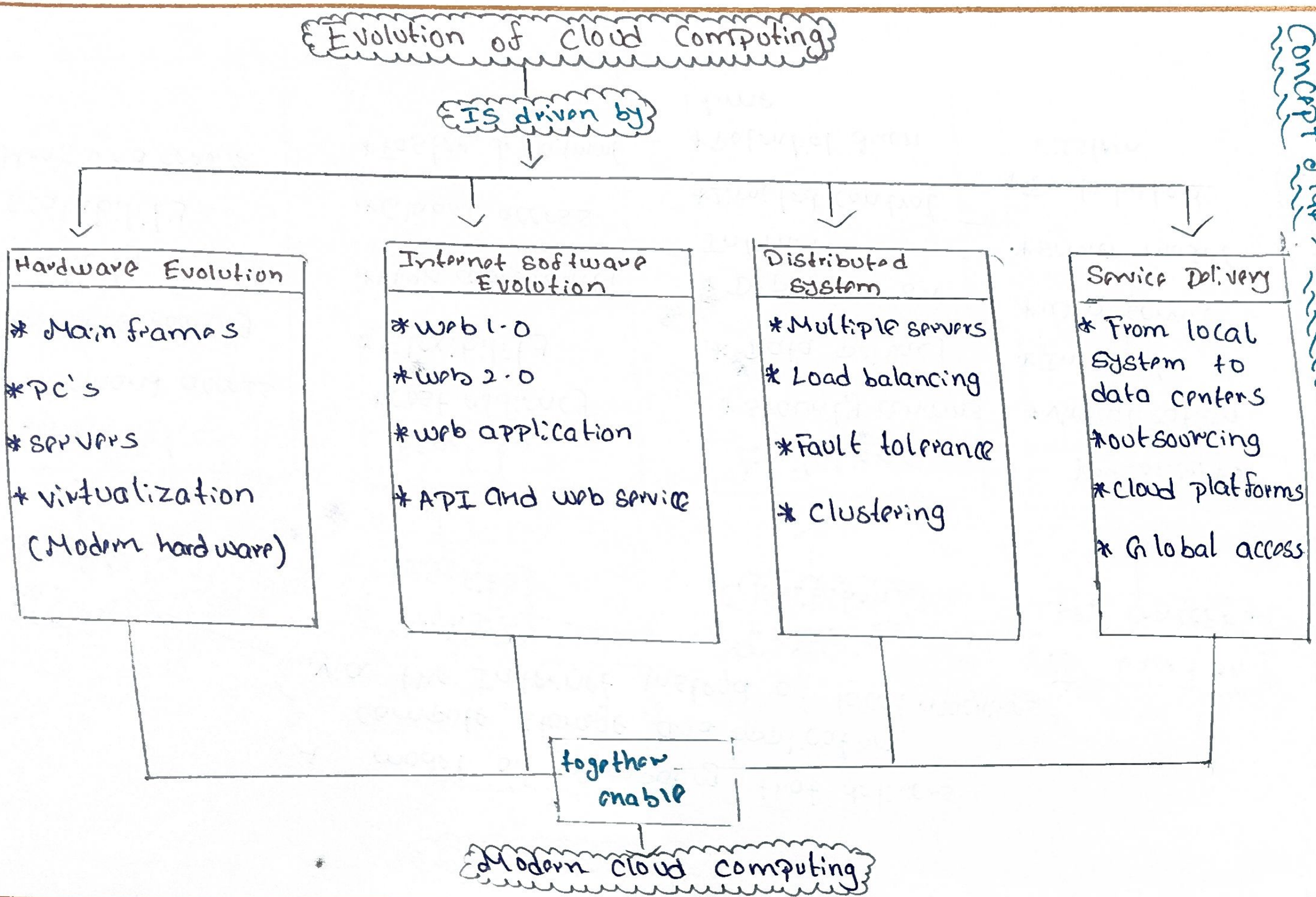
Limitations

* security concerns
* Data privacy
* Dependance on Internet
* Limited control
* Potential down time

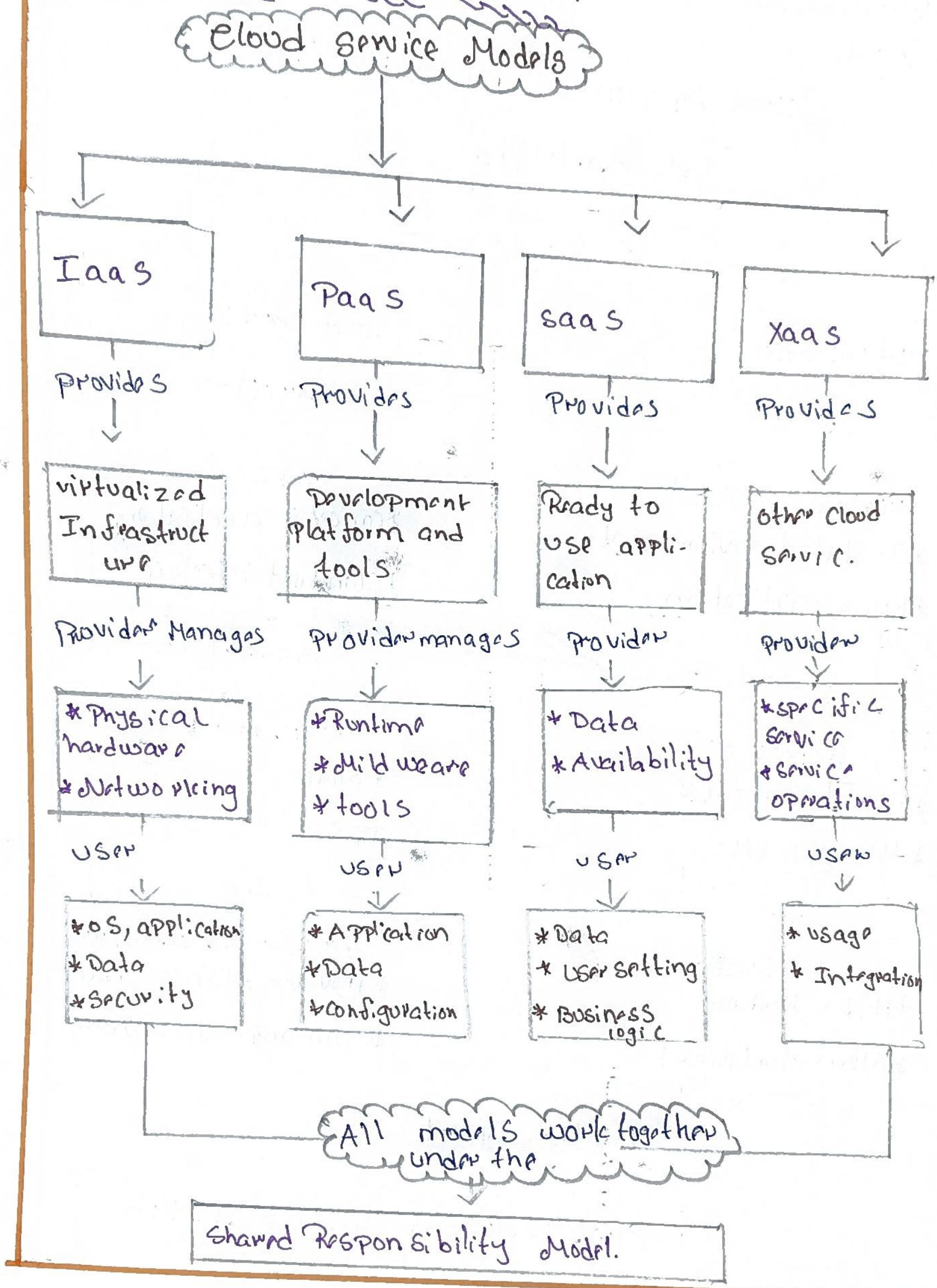
Key concepts

* virtualization
* Internet
* web service
* service model
* Distributed system.

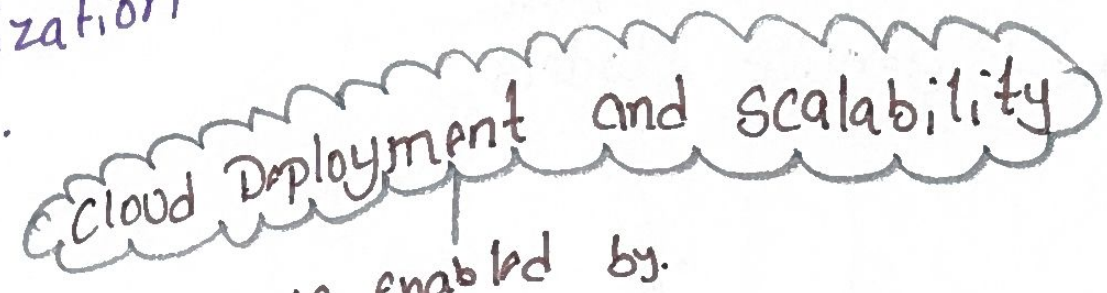
Concept Map → Evolution of Cloud Computing :-



Concept Map:- Service Models:-



4) virtualization and web service as Enablers of cloud.



are enabled by.

virtualization

Provides

virtual Machine
*Isolated environment
*Runs application

managed by

Hypervisor
*Allocates resource
*Manages VMs

uses

Abstraction
*Hides hardware
*Allows portability

web service

Provides

API
*Concepts applications
*Standard interface

managed by

SOAP / Rest
*SOAP
*REST

uses

Provider and consumer
*Provider offers service
*consumer uses service

together support

Cloud Deployment and Scalability